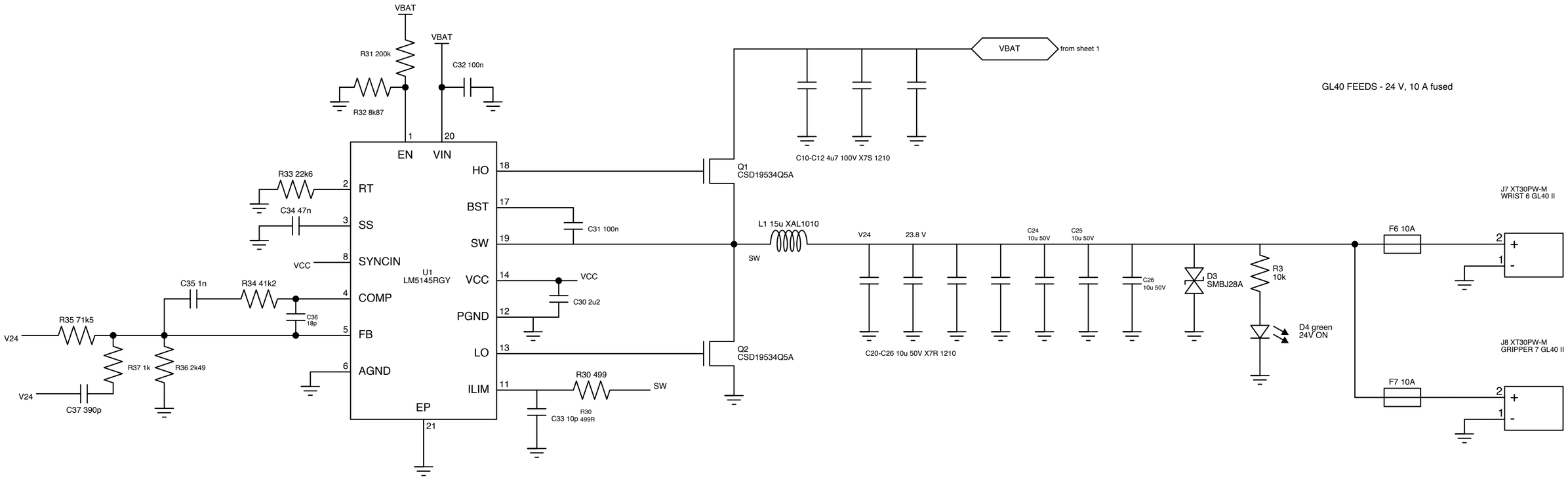
24 V BUCK - TI LM5145 (SNVSAI4B, application circuit 3: 48 V -> 24 V, 440 kHz, 5 A cont.)



NOTES

1. Values exactly as TI's application circuit 3 (compensation is TI's); RFB1 71k5 (E96) sets 23.8 V.

2. SYNCIN tied to VCC: forced PWM, so GL40 regen flows back to VBAT instead of raising V24.

3. 5 A continuous; the valley limit (R30 on Q2's RDS(on)) lets ~8-9 A out for peaks.

4. UVLO: R31 / R32 start the converter at 28.3 V, stop it at 26.3 V.

5. J7 / J8 are XT30 male: a GL40 lead cannot reach a 49 V socket, nor an AK lead a 24 V one.

24 V CONVERTER + GL40 FEEDS	
PIONEER-ARM-PDB (cad_bench/projects/arm_pdb.py)	
TI LM5145 application circuit 3, unchanged	
SHEET 2 / 2	REV A
DATE 2026-11-05	SIZE A3